



OPEN Effects of knowledge and importance on responsibility in human-AI decision making

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As agents such as AI systems and robots increasingly support human decision-making, questions of accountability in cases of failure have become critical. Prior research has examined responsibility attribution mainly in terms of system autonomy, transparency, or anthropomorphism, but little is known about how cognitive framing (prior knowledge of agents) and contextual framing (perceived importance of a task) jointly shape these judgments. This study addresses this gap through a three-factor mixed-design experiment with 588 participants. Participants evaluated responsibility for the user, the agent, and the agent's developer or provider after observing failed agent-assisted interactions. The results showed that prior knowledge of the agent shifted responsibility away from the user and toward the agent and its developer. Moreover, when the topic was perceived as highly important, responsibility attributed to the developer or provider increased substantially. These findings highlight that responsibility attribution in human-agent interaction is dynamic rather than static, modulated by both user expectations and situational seriousness. Beyond theoretical contribution, the results suggest practical implications for system design, user education, and legal policy, offering guidance on how to reduce accountability gaps in the deployment of socially embedded agents.

Keywords Human-agent interaction (HAI), Human-robot interaction (HRI), Responsibility, Empathy agent

Humans increasingly rely on artificial intelligence (AI) systems and robots—hereafter collectively referred to as **agents**, reflecting their growing role as semi-autonomous decision-making entities—to support a variety of tasks in daily life. Popular technologies such as ChatGPT, autonomous driving systems, and industrial robots have become widespread across the globe. However, the deeper integration of agents into society has raised serious concerns related to trust, ethics, and accountability in human-agent interaction. Prior studies on accountability in AI have often overlapped with the literature on explainability and transparency, emphasizing how much information systems should disclose to ensure fair use^{1,2}. However, explanation alone has been shown to be insufficient in addressing responsibility, particularly when users do not accept explanations or when explanations are unavailable^{3,4}. This limitation highlights the need to examine responsibility attribution beyond explanation-centric approaches.

Kaplan et al⁵ identified three primary categories—human characteristics, AI performance, and contextual challenges—that influence trust in AI. In contrast, Munn⁶ criticized existing AI ethics principles as largely ineffective in mitigating real-world harms, arguing that systemic power dynamics must be addressed to ensure equitable outcomes. These perspectives underscore that trust and accountability are shaped not only by system transparency but also by broader social and contextual factors.

As agents are increasingly entrusted with supporting human decision-making, the issue of responsibility attribution has become crucial. Failures in domains such as healthcare, autonomous driving, and legal advice can lead to significant harm, making it necessary to determine who should be held accountable—the user, the agent, or its developer. Alické's Culpable Control Model⁷ suggests that people often rely on emotion and intuition when assigning blame. Similarly, Malle et al⁸ demonstrated that blame involves moral evaluation, intent attribution, and norm assessment, indicating that judgments toward agents depend not only on outcomes but also on perceived mental states and agency.

This becomes especially relevant when agents exhibit autonomy or human-like features. According to the media equation theory⁹, users apply social heuristics to interactive systems. Waytz et al¹⁰ showed that anthropomorphizing autonomous vehicles increases trust and alters blame attribution. Furlough et al¹¹ found that highly autonomous agents receive more blame than low-autonomy ones, while Joo¹² demonstrated that

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attributing a human-like mind to AI increases blame toward the system and decreases blame toward humans, raising concerns about moral scapegoating.

Beyond anthropomorphism and autonomy, recent experimental work has explored responsibility allocation in concrete interaction scenarios. Stojilović et al.¹³ reported that autonomous vehicles are consistently blamed more than human drivers, while Franklin et al.¹⁴ highlighted the multifactorial nature of responsibility judgments. These findings suggest that blame is influenced not only by system design but also by how the situation is framed. Yet, such work has not systematically examined user-related cognitive factors together with contextual importance. However, not all users accept or trust anthropomorphized systems. Nomura et al.^{15,16} documented user resistance to robots, and empirical studies by Tsumura and colleagues^{17–23} found that while people can empathize with agents, the intensity of trust or empathy is often weaker than toward humans. These results illustrate the tension between anthropomorphic design and accountability expectations.

In addition to agent appearance and autonomy, system transparency plays a key role in trust and responsibility attribution. Okamura and Yamada²⁴ emphasized trust calibration—ensuring that users neither over-trust nor under-trust AI systems. Apologies and explanations have also been shown to influence blame and trust repair²⁵. These psychological and design factors intersect with normative debates on AI responsibility. Some scholars²⁶ argue that courts may soon need to consider whether agents qualify as legal actors, while others warn against granting legal personhood²⁷. Santoni de Sio and van den Hoven²⁸ proposed “meaningful human control” as a principle to prevent responsibility gaps by ensuring that humans remain involved in critical decisions.

Despite substantial prior work, the combined influence of user cognition (e.g., prior knowledge of the system) and contextual seriousness on responsibility attribution remains insufficiently understood. These factors may dynamically shape how people distribute blame when agent-assisted decisions fail, yet they have received limited empirical attention. In particular, little is known about how users’ prior knowledge of agents and the perceived importance of a task jointly influence responsibility judgments when failures occur. This study addresses this gap by examining how these cognitive and contextual factors shape blame attribution across three targets: the user, the agent, and the agent’s developer or service provider. Understanding these patterns is theoretically important for models of blame attribution and practically relevant for system design, legal policy, and public trust. Clarifying responsibility in human–agent collaboration can reduce accountability gaps and support ethical deployment, especially in high-stakes domains such as healthcare, disaster response, and autonomous mobility.

Related work

Responsibility attribution in human-agent interaction has been studied across disciplines including psychology, law, and human-robot interaction (HRI). Across these domains, prior work has highlighted factors such as causal structure, system autonomy, and moral intuition as central to blame judgments. In the context of automated vehicles (AVs), Stojilović et al.¹³ found that people attribute greater responsibility when AVs are involved in accidents, compared with scenarios with human drivers. Franklin et al.¹⁴ proposed a comprehensive framework comprising nine factors—such as causality, knowledge, autonomy, and intention—that influence responsibility judgments. They also introduced serious games as tools to explore responsibility in causal cognition.

Medical and legal contexts provide further insight into responsibility attribution. Naik et al.²⁹ reviewed how responsibility is assigned when AI is used in clinical decision-making, while Parlange et al.³⁰ explored how delegating legal decisions to AI alters moral judgments. Scheutz and Malle³¹ examined moral evaluations of agents making life-or-death decisions and found that such decisions are judged through human moral lenses. Kneer and Stuart³² reported that people tend to hold AI systems responsible in reckless scenarios.

From a normative perspective, Ryan³³ warned against overly trusting AI systems, cautioning against anthropomorphism. Hagendorff³⁴ reviewed 22 AI ethics guidelines and identified overlapping themes as well as notable gaps. Coeckelbergh³⁵ and de Jong³⁶ highlighted the challenges of attributing liability in complex, multi-actor systems. Maas³⁷ emphasized the destabilizing effect of AI on existing governance frameworks, while Danaher³⁸ argued that unchecked AI deployment could erode society’s moral foundations. Regarding the philosophical basis of agency, Nyholm³⁹ examined the extent to which autonomous systems like self-driving cars or military robots can be considered action-subjects. Danaher⁴⁰ introduced the concept of the “retaliation gap,” showing that traditional frameworks struggle to accommodate blame in AI-related harms.

From moral psychology, dual-process models of blame suggest that people weigh both intentions and outcomes when evaluating harmful actions. Cushman⁴¹ emphasized that intentionality plays a key role in moral evaluations, particularly when an outcome could have been foreseen or prevented. Extending this work to AI, Hidalgo et al.⁴² found that algorithmic agents are often blamed more for physical harms, while humans are blamed more for violations of fairness. Furlough et al.⁴³ showed that when task failures occur in human-robot teams, autonomous robots are assigned more blame than their non-autonomous counterparts, indicating that perceived autonomy enhances responsibility attribution.

Trust calibration also plays a critical role in how people assign responsibility to agents. de Visser et al.⁴³ found that anthropomorphic design can enhance trust resilience, while a lack of transparency may lead to overblaming the AI. Recent work by Bai and Chen⁴⁴ demonstrated that different explanation strategies from robots—such as external attributions or admitting low ability—have varying effects on trust repair. Their findings suggest that attributing failure to external causes was more effective in restoring trust than internal attributions involving low integrity, which significantly harmed trust. These results highlight the nuanced impact of how agents explain their failures and the importance of aligning explanation strategies with user expectations to foster appropriate responsibility attribution.

The degree of anthropomorphism also affects how blame is distributed. Joo⁴² found that when people perceive AI as having a human-like mind, they tend to assign greater blame to the AI and less to the human counterpart, sometimes leading to moral scapegoating. Gall and Stanton⁴⁵ demonstrated that people shared more accountability with human-like robotic teammates, whereas machine-like agents were blamed more

heavily. Together, these findings suggest that design cues influencing perceived agency or social presence can substantially alter moral and legal judgments.

From a design ethics perspective, Value Sensitive Design (VSD)⁴⁶ was proposed; it embeds mechanisms for transparency, traceability, and shared responsibility directly into AI systems. This includes audit trails, user-facing explanations, and confirmation mechanisms for critical decisions. These practices aim to reduce responsibility gaps and promote meaningful human control, aligning system behavior with stakeholder values. Finally, the concept of the “responsibility gap”⁴⁷ is often cited in discussions about autonomous systems. As AI agents gain autonomy, it becomes difficult to trace clear lines of liability. Although some policymakers have proposed granting AI legal personhood, many scholars argue that this would obscure human accountability²⁷. Most agree that moral and legal responsibility must remain with the humans who design, deploy, or oversee these systems. Yet, despite this growing body of work, little is known about how users’ prior knowledge of agents and the perceived importance of tasks jointly shape responsibility judgments. Addressing this gap is the central aim of the present study.

Methods

Ethics statement

The protocol was approved by the ethics committee of the National Institute of Informatics (No. R6-11-1,11, July, 2024) and Toyo university (No. 2024-001,17, April, 2024). All studies were carried out in accordance with the recommendations of the Ethical Guidelines for Medical and Health Research Involving Human Subjects provided by the Ministry of Education, Culture, Sports, Science and Technology and Ministry of Health, Labour and Welfare in Japan. Written informed consent was provided by choosing one option on an online form: “I am indicating that I have read the information in the instructions for participating in this research. I consent to participate in this research.” All participants gave informed consent. After that, they were debriefed about the experimental procedures. The experiment was conducted from 27 to 28 January 2025 (Japan time).

Hypotheses

The aim of this study is to examine how responsibility is attributed when an agent fails in assisting a human user. Specifically, we investigated the extent to which participants assign blame to three possible targets: (1) the user, (2) the agent, and (3) the agent’s developer or service provider. To explore these attribution patterns, we formulated two hypotheses based on prior literature.

H1: Prior knowledge of the agent increases perceived accountability of both the agent and its developer or provider.

H2: The more important the topic is to the participant, the more responsibility is attributed to the agent’s developer or provider (and relatively less to the user).

H1 draws on research on expectation and agency attribution^{13,48}. Prior knowledge of a system shapes user expectations of its performance; when these expectations are violated, people tend to ascribe greater responsibility to the system and those behind it. This prediction assumes—consistent with prior work on human–human blame attribution^{7,8}—that the psychological processes underlying responsibility judgments (such as expectation violation and norm evaluation) may extend to evaluations of artificial agents. Because these mechanisms have been established primarily in human–human contexts, we treat their application to human–agent scenarios as a theoretical extension rather than an established fact. We also assume that observers can form meaningful responsibility judgments even when evaluating interactions vicariously. Accordingly, we predict that familiarity with the agent will increase responsibility attributed to both the agent and its developer or provider.

H2 is grounded in studies on perceived seriousness and delegation of judgment^{30,32}. When the stakes of a task are high, individuals are less willing to blame themselves and more inclined to hold responsible those perceived to have ultimate control or expertise. This hypothesis similarly assumes—based on prior findings in human–human moral and blame judgments^{7,8}—that the cognitive heuristics guiding responsibility allocation may also influence judgments involving artificial agents. Although evidence remains limited, recent work suggests that certain aspects of moral judgment can extend to non-human actors¹². This extension, however, should be viewed as a theoretical assumption rather than an established empirical fact. Thus, we predict that higher perceived importance of the topic will increase responsibility attributed to the agent’s developer or provider, while decreasing responsibility attributed to the user.

Experimental procedure

The experiment involved exposing participants to short videos that depicted a dialogue between a human user and an artificial agent designed to assist with decision-making. In each video, the agent attempted to provide support but ultimately failed to meet the user’s needs—for instance, by offering an inappropriate recommendation, by delivering information that was incomplete or irrelevant, or by misunderstanding the user’s request. To minimize the influence of paralinguistic cues such as tone or intonation, all videos were presented **without audio narration**. The entire interaction was conveyed through on-screen text, ensuring that participants based their evaluations solely on the content of the dialogue rather than on voice-related factors. After watching the assigned video(s), participants were asked to evaluate the outcome by completing a short questionnaire designed to capture how responsibility was attributed to different parties.

Presenting the interaction without audio narration was intended to avoid well-documented voice-related biases, such as shifts in perceived warmth, competence, trust, and agency that arise from paralinguistic cues^{49,50}. However, voice is not the only source of potential bias. Differences in participants’ reading speed or linguistic fluency could also influence comprehension of on-screen text. To minimize this concern, all dialogue was

written in simple, commonly used Japanese, and participants were allowed unlimited time to read each segment. Prior studies indicate that online crowdsourcing populations in Japan generally possess sufficient literacy for text-based experimental tasks^{17,51}. Thus, while removing audio controlled for voice-related biases, the study also took steps to reduce comprehension-related variance across participants.

The study was conducted online using Yahoo! Crowdsourcing, which has been frequently employed in previous experimental research to obtain reliable and diverse samples under controlled conditions^{17,19,20,52}. Participants could take part from their own devices, and instructions emphasized that they should carefully imagine themselves as the user in the video. This self-referential framing was intended to elicit natural responsibility judgments rather than detached third-person evaluations. A flowchart of the overall procedure is provided in Fig. 1.

The experiment adopted a three-factor mixed design. The between-participants factors were:

- **Prior knowledge of the agent** (yes vs. no), manipulated by showing or withholding an introductory video that described the agent's intended functions, capabilities, and everyday applications.
- **Importance of the topic** in the user-agent dialogue (low, medium, high), manipulated by varying the stakes of the scenario. Low-importance cases involved trivial or everyday decisions, medium cases reflected moderately consequential topics, and high-importance cases concerned situations with serious personal or social consequences. To illustrate these levels in real-world terms, the low-importance scenario corresponded to a minor everyday situation (e.g., receiving a job-related notification), the medium-importance scenario involved a moderately consequential choice (e.g., selecting a travel destination), and the high-importance scenario reflected a high-stakes context with potentially severe outcomes (e.g., responding to an emergency).
- The within-participants factor was the **responsibility target** (user, agent, developer/service provider), which was assessed through participants' ratings.

In practice, participants assigned to the prior knowledge condition first watched a short introductory video (also silent with on-screen text) that presented the agent's role and capabilities. All participants then proceeded to the main task video, in which the user-agent dialogue was shown and the agent's assistance led to an unsatisfactory outcome. The scenario differed according to the assigned level of topic importance, thereby allowing us to examine whether the perceived gravity of the situation altered responsibility judgments. After completing the viewing, participants responded to the responsibility questionnaire, which asked them to evaluate the extent to which each of the three targets was responsible for the outcome.

In total, the design comprised 18 experimental conditions (2 levels of prior knowledge $\times$ 3 levels of topic importance $\times$ 3 responsibility targets). Because responsibility target was treated as a within-participants factor, each participant was assigned to one of six experimental groups, each corresponding to a unique combination of the between-participants factors. This design allowed us to capture both the contextual influences of knowledge and importance as well as the comparative judgments across different responsibility targets.

Participants

Participants were recruited through Yahoo! Crowdsourcing and received a monetary reward of 55 yen for their participation. A dedicated web page was created for the experiment, and the user and agent characters in

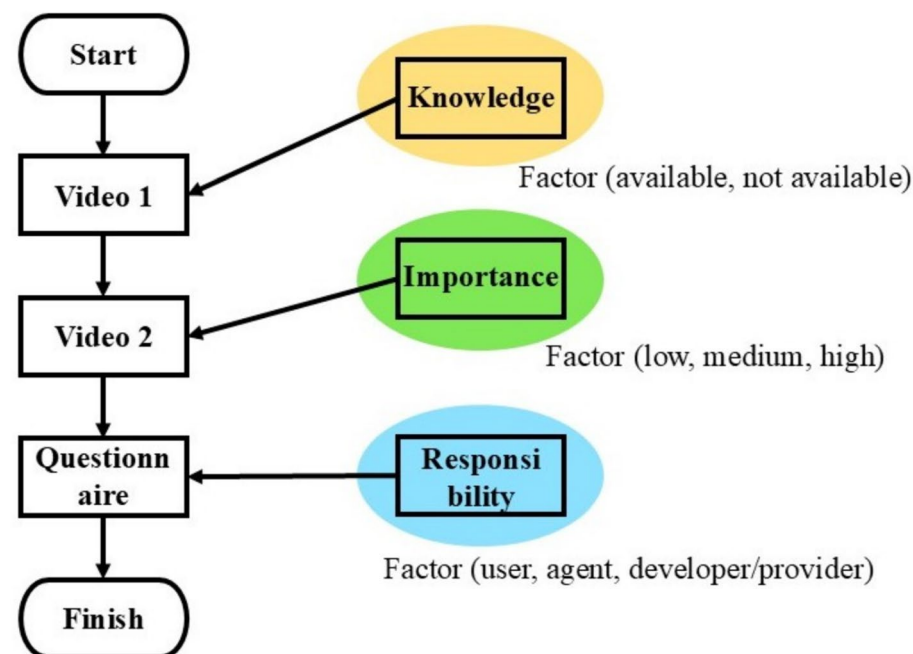


Fig. 1. Flowchart of the experimental procedure.

the videos were animated using MikuMikuDance to ensure standardized and reproducible stimuli. A total of 614 individuals initially took part in the study. To ensure data quality, participants were asked content-check questions immediately after viewing the videos. Fourteen participants who failed these checks were excluded from the dataset.

To maintain a balanced distribution across conditions, we included the first 98 valid participants in each of the six experimental groups, yielding a final sample of 588 participants. This procedure ensured that each between-participants condition was represented by an equal number of participants, enhancing the comparability of results across experimental manipulations. The final sample had a mean age of 49.88 years ($SD = 12.28$), with ages ranging from 19 to 89. The gender distribution was nearly balanced, with 297 males and 291 females, providing a demographically diverse participant pool.

Questionnaire

To assess the extent to which participants attributed responsibility for the outcome of the user–agent interaction, we developed a brief questionnaire consisting of three items. Each item corresponded to one of the three responsibility targets: the user, the agent, and the agent’s developer or service provider. Specifically, participants were asked:

- Q1: To what extent was the user responsible?
- Q2: To what extent was the agent responsible?
- Q3: To what extent was the agent’s developer or service provider responsible?

Responses were collected on a 10-point Likert scale, ranging from 1 (no responsibility) to 10 (full responsibility). This scale format was adapted from the 100-point responsibility attribution scale used by Stojilović et al.¹³, but was deliberately simplified to enhance clarity, reduce participant fatigue, and ensure efficient responding in an online crowdsourcing environment. Specifically, the original 0–100 scale was converted into a 1–10 format by compressing the range into ten evenly spaced categories, each representing a 10-point interval on the original scale (e.g., 0–10 → 1, 11–20 → 2, ..., 91–100 → 10). No semantic labels were changed; only the numeric granularity was reduced to match the shorter and more intuitive format for online participants.

Participants completed the questionnaire immediately after watching the video and were explicitly instructed to imagine themselves in the role of the user. This framing encouraged first-person responsibility judgments rather than detached third-person evaluations, thereby enhancing the ecological validity of the responses. The order of the three responsibility questions—first the user, followed by the agent, and finally the developer—was fixed for all participants; the items were not counterbalanced.

Prior knowledge of the agent

To examine whether prior knowledge of the agent influenced participants’ responsibility attribution, we created a short self-introduction video (approximately 90 seconds) in which the agent explained its role and characteristics. Participants assigned to the prior knowledge condition were required to watch this video (Video 1), which consisted of a silent presentation with on-screen text describing the types of assistance the agent could provide, as well as details about its personality traits and behavioral style. The purpose of this manipulation was to ensure that participants in this condition had a more concrete impression of the agent as a social and functional entity before proceeding to the main task.

In contrast, participants in the no-prior-knowledge condition were shown only a still image of the agent without any additional contextual information. This design allowed us to compare responsibility judgments between participants who had prior exposure to the agent’s self-presentation and those encountering the agent for the first time during the main task. Figure 2 illustrates the agent’s visual appearance and provides an example frame from the self-introduction video. By combining both the static representation and a sample video frame into a single figure, we convey the essential features of the stimulus while keeping the total number of figures within the journal’s constraints.

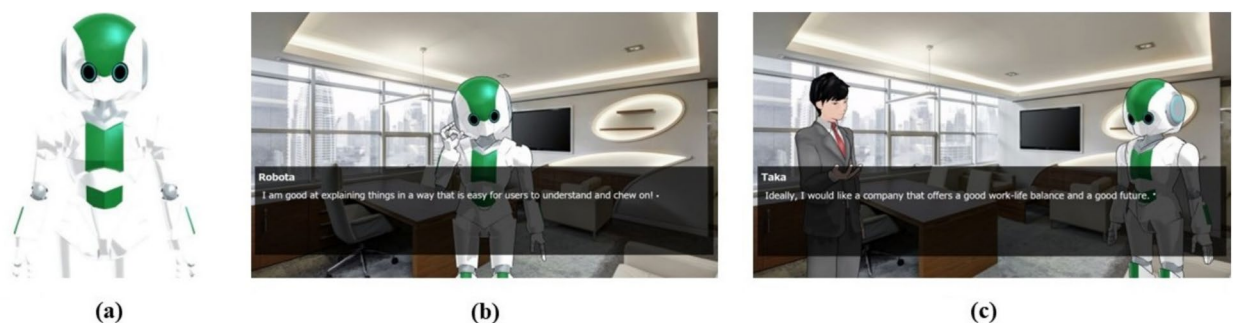


Fig. 2. Examples of experimental stimuli: (a) visual appearance of the agent, (b) a frame from the self-introduction video, and (c) a frame from the user–agent interaction in the main task.

Importance of the topic

To examine how the perceived importance of a situation affects responsibility attribution, we prepared three different dialogue videos between the user and the agent, each representing a different level of topic importance. The levels of importance were established through a pretest in which a separate group of participants rated the perceived seriousness of a range of scenarios. Based on these ratings, we selected three representative categories:

- **Low importance:** everyday situations such as job opportunities that involve limited personal stakes.
- **Medium importance:** moderately consequential contexts such as choosing travel destinations, where the decision has some impact but remains reversible.
- **High importance:** highly consequential contexts such as emergency situations, in which inappropriate decisions may result in severe negative outcomes.

These scenarios were incorporated into the main experimental video (Video 2), with each participant viewing only one version. Each video lasted approximately 120 seconds. To minimize confounding variables, the gestures, facial expressions, and turn-taking between the user and the agent were standardized across all versions. All videos were presented without audio, ensuring that participants’ evaluations were not influenced by voice intonation or other paralinguistic cues. In the videos, the male character on the left (Taka) served as the user, while the robot character on the right (Robota) represented the agent. The developer or service provider of the agent was not depicted in any of the scenarios, thereby ensuring that participants’ judgments about this target were based on inference rather than direct portrayal. Example frames of the agent’s appearance, its self-introduction video, and the main task conversation are presented together in Fig. 2.

Manipulation check

To validate whether participants perceived differences in the importance of various topics, we conducted a preliminary survey with 102 participants (47 males and 55 females). The mean age was 49.18 years (*SD* = 12.17), with a range from 20 to 89 years. Participants were asked to evaluate nine topics that could plausibly be discussed with an agent. Each topic was rated on a 7-point Likert scale (1 = very low importance, 7 = very high importance). As all participants evaluated all topics, topic importance was treated as a within-participants factor.

A one-way repeated measures ANOVA revealed a significant main effect of topic on perceived importance, $F(5.98, 604.33) = 16.40, p < .001$. To account for violations of sphericity, Mendoza’s test was applied, yielding a Greenhouse–Geisser epsilon (ϵ) of 0.7479, and the degrees of freedom were adjusted accordingly. As shown in Table 1, mean importance ratings varied across topics. Holm-adjusted pairwise comparisons indicated that employment opportunities were perceived as significantly less important, planning a trip represented a moderate level of importance, and emergency situations were rated as significantly more important. On the basis of these results, we selected these three topics to represent the low, medium, and high importance conditions in the main experimental videos.

Analysis method

An analysis of variance (ANOVA) was conducted using a three-factor mixed design, which is commonly used in previous studies and is suitable for the present research. The between-participants factors included two levels of prior knowledge of the agent and three levels of topic importance. The within-participants factor was the responsibility target, with three levels (user, agent, and agent’s developer or provider). Responsibility ratings obtained from the questionnaire were used as the dependent variable to examine how blame attribution varied across conditions when agent-supported decisions failed. All statistical analyses, including ANOVA and multiple comparisons, were performed using R (version 4.1.0), which was also used for the manipulation check.

Before conducting the main analyses, we examined the statistical assumptions required for mixed-design ANOVA. For the within-participants factor (responsibility target), sphericity was tested using Mendoza’s test.

Topic	Mean	SD
T1: Hobbies and pastimes	4.40	1.37
T2: Regional issues	4.18	1.23
T3: AI medical decisions	4.74	1.20
T4: Planning a trip	4.44	1.32
T5: Educational policy	3.77	1.38
T6: Emergency situations	4.99	1.03
T7: Shopping	4.68	1.28
T8: Employment opportunities	3.74	1.40
T9: Legal and rights issues	4.59	1.33

Pairwise comparison	Difference	t	p
T6 (Emergency) – T8 (Employment)	1.26	8.80	< .001 ***
T4 (Trip) – T8 (Employment)	0.71	4.24	< .01 *
T4 (Trip) – T6 (Emergency)	–0.55	3.85	< .01 *

Table 1. Perceived importance of topics (top) and results of pairwise comparisons (bottom). *p*: * *p* < 0.05, *** *p* < 0.001.

Knowledge	Importance	Responsibility	Mean	S.D.	Knowledge	Importance	Responsibility	Mean	S.D.
Available	Low	User	5.878	2.042	Not available	Low	User	6.153	2.400
		Agent	6.122	1.890			Agent	5.500	2.184
		Developer/provider	6.489	1.960			Developer/provider	6.143	2.346
	Medium	User	5.480	2.321		Medium	User	5.980	2.377
		Agent	5.990	2.098			Agent	5.633	2.175
		Developer/provider	6.439	2.351			Developer/provider	6.031	2.477
	High	User	4.643	2.017		High	User	5.276	2.219
		Agent	6.592	1.978			Agent	6.398	2.157
		Developer/provider	7.204	1.921			Developer/provider	7.031	2.073

Table 2. Statistical information on responsibility targets.

Factor	df	F	p	η_p^2
Prior knowledge of agent (Knowledge)	1	0.4906	0.4840 <i>ns</i>	0.0008
Importance of topic (Importance)	2	1.940	0.1446 <i>ns</i>	0.0066
Recipient of responsibility (Responsibility)	1.45	32.56	0.0000 ***	0.0530
Knowledge $\times$ Importance	2	0.7057	0.4942 <i>ns</i>	0.0024
Knowledge $\times$ Responsibility	1.45	7.520	0.0021 **	0.0128
Importance $\times$ Responsibility	2.9	14.20	0.0000 ***	0.0465
Knowledge $\times$ Importance $\times$ Responsibility	2.9	0.0958	0.9589 <i>ns</i>	0.0003

Table 3. Analysis results of ANOVA. *p*: ***p*<0.01 ****p*<0.001.

When violations of sphericity were detected, Greenhouse–Geisser epsilon values were computed, and the degrees of freedom for the within-participants effects and their interactions were adjusted accordingly. All ANOVA results reported in this article reflect these corrected degrees of freedom. In addition, Holm-adjusted pairwise comparisons were conducted for follow-up tests, ensuring control of the familywise error rate while maintaining statistical power. These procedures are widely recommended for mixed-model ANOVA designs and align with standard analytical practices in HAI and social cognition research.

Results

Table 2 presents the means and standard deviations of responsibility ratings across all experimental conditions. In what follows, we first report descriptive statistics and then present the results of the three-factor mixed-design ANOVA.

The ANOVA results are summarized in Table 3. Significant interactions were observed between prior knowledge and responsibility target, and between topic importance and responsibility target. Given these interactions, the interpretation of main effects alone would be misleading; therefore, we focus on the results of simple main effects analyses, presented in Table 4.

For the knowledge $\times$ responsibility interaction, participants with prior knowledge of the agent attributed greater responsibility to both the user and the agent compared to those without prior knowledge, whereas responsibility attributed to the developer or provider did not vary significantly across conditions. For the importance $\times$ responsibility interaction, responsibility attributed to the developer or provider increased markedly as topic importance rose, while user responsibility decreased accordingly. Responsibility attributed to the agent remained moderate across all conditions, but also showed an upward trend under high-importance scenarios.

Taken together, these results provide empirical support for our hypotheses. Specifically, H1 was partially supported, as prior knowledge increased perceived accountability of both the agent and its developer or provider. H2 was also supported, as greater topic importance led to increased responsibility attributed to the developer or provider and decreased responsibility attributed to the user.

Knowledge $\times$ responsibility

The interaction between prior knowledge of the agent and responsibility attribution was significant (Table 4), and the detailed distributions are illustrated in Fig. 3. Figure 3(a) shows the results for responsibility attributed to the user. Participants without prior knowledge of the agent assigned significantly greater responsibility to the user compared with those who had prior knowledge. Figure 3(b) presents the results for responsibility attributed to the agent. In contrast, participants with prior knowledge of the agent were more likely to attribute responsibility to the agent itself, relative to those without such knowledge.

Figure 3(c) and (d) illustrate the distribution of responsibility ratings across all three targets under the prior knowledge and no-prior-knowledge conditions, respectively. In the prior knowledge condition (Fig. 3c), responsibility was attributed highest to the developer or service provider, followed by the agent, and lowest to

	Factor	df	F	p	η_p^2
Knowledge $\times$ Responsibility	Knowledge at user	1	6.487	0.0111 *	0.0110
	Knowledge at agent	1	5.182	0.0232 *	0.0088
	Knowledge at developer/provider	1	2.915	0.0883 <i>ns</i>	0.0050
	Responsibility at available knowledge	1.4	36.35	0.0000 ***	0.1110
	Responsibility at no available knowledge	1.49	6.764	0.0035 **	0.0227
Importance $\times$ Responsibility	Importance at user	2	11.72	0.0000 ***	0.0387
	Importance at agent	2	7.036	0.0010 ***	0.0236
	Importance at developer/provider	2	9.649	0.0001 ***	0.0321
	Responsibility at low importance	1.43	2.832	0.0781 <i>ns</i>	0.0144
	Responsibility at medium importance	1.48	2.937	0.0703 <i>ns</i>	0.0149
	Responsibility at high importance	1.42	62.70	0.0000 ***	0.2443

Table 4. Analysis results of simple main effect. * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

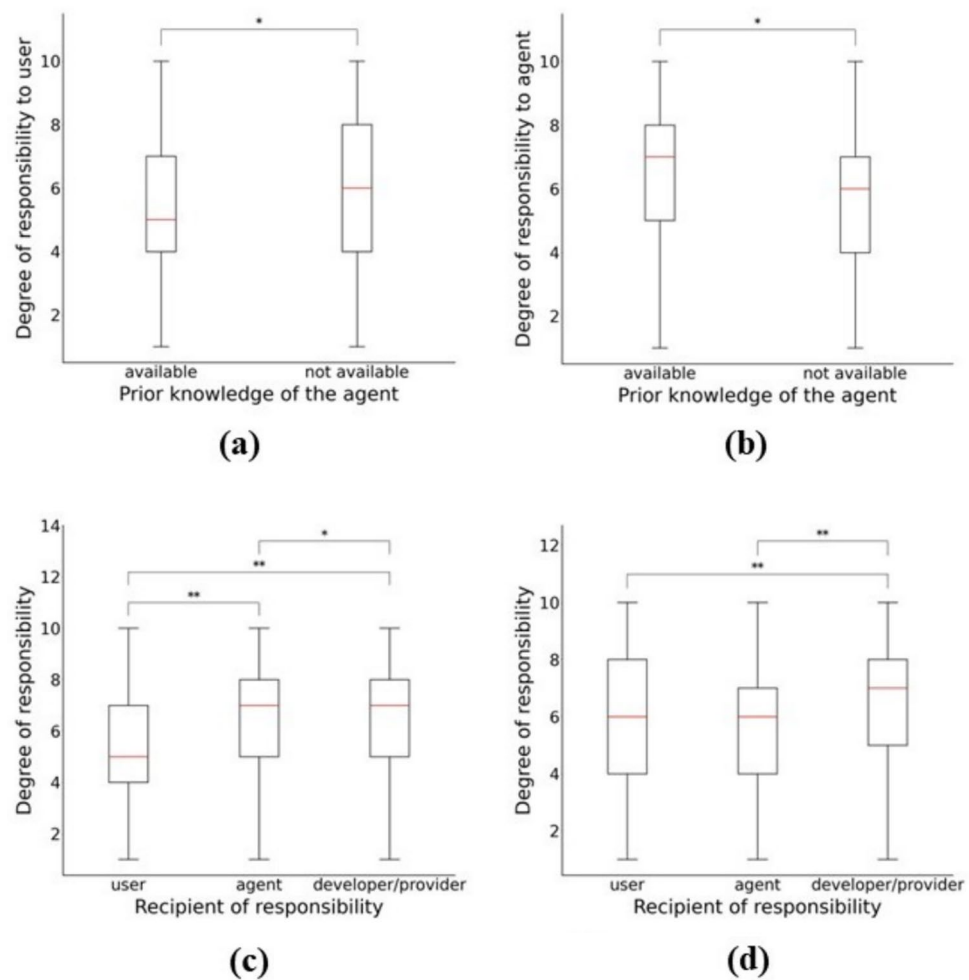


Fig. 3. Results of Knowledge $\times$ Responsibility interaction. Red lines indicate medians; circles represent outliers.

the user. In the no-prior-knowledge condition (Fig. 3d), responsibility was still rated highest for the developer or service provider, while the agent and the user were attributed similar, lower levels of responsibility.

Importance $\times$ responsibility

A significant interaction between topic importance and responsibility attribution was observed (Table 4), and the detailed patterns are illustrated in Fig. 4. Figure 4(a) presents the results for responsibility attributed to the

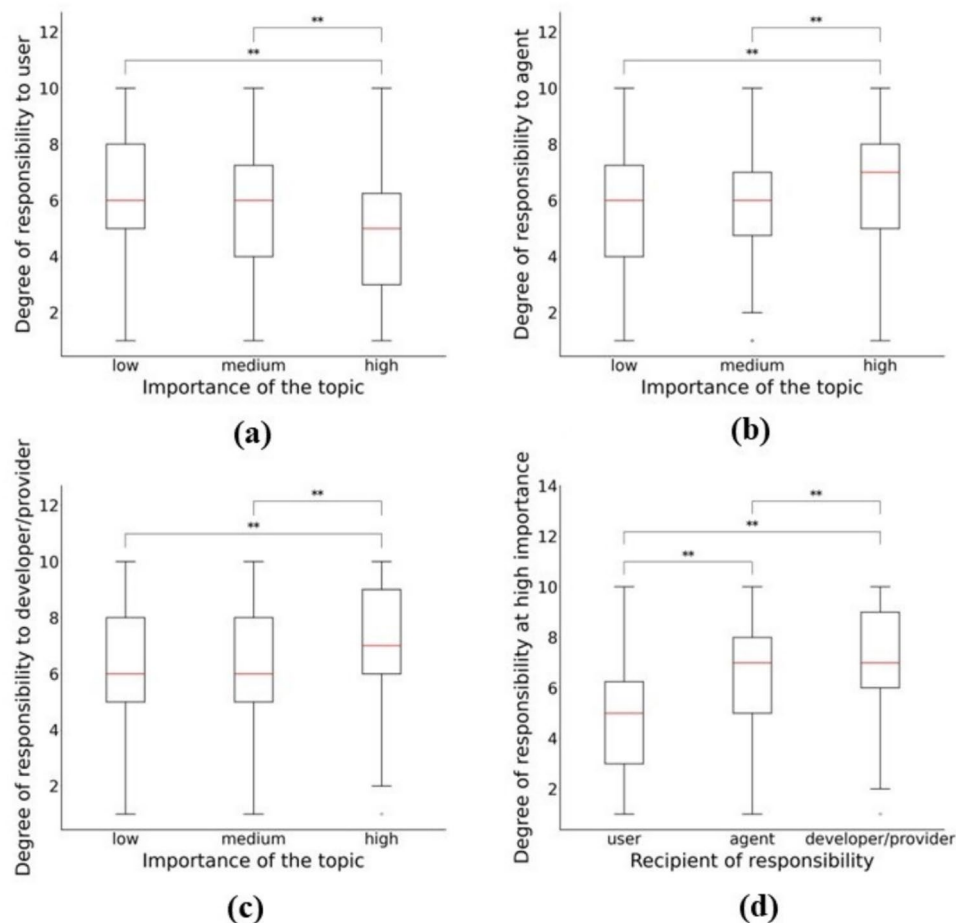


Fig. 4. Results of Importance $\times$ Responsibility interaction. Red lines indicate medians; circles represent outliers.

user. Responsibility ratings for the user decreased as topic importance increased. Figure 4(b) shows the results for the agent. In contrast, responsibility attributed to the agent increased progressively with higher levels of topic importance.

Figure 4(c) illustrates the results for the agent's developer or service provider. Responsibility ratings also increased as topic importance rose, indicating that more serious scenarios prompted greater accountability directed at the system's creators. Finally, Fig. 4(d) displays the distribution of responsibility ratings in the high-importance condition. Responsibility was highest for the developer or service provider, followed by the agent, and lowest for the user, demonstrating a systematic upward shift of accountability as the stakes of the situation increased.

Discussion

Supporting hypotheses

Failures can occur even when agents are carefully designed to assist users. Understanding how responsibility attribution shifts depending on contextual factors is critical for improving both the design and the social acceptance of such systems. If we can clarify how responsibility is distributed across users, agents, and developers in different situations, designers and policymakers can better anticipate accountability gaps and implement strategies to prevent undue blame being placed on any single party. This study examined whether two key factors—(1) prior knowledge of the agent and (2) the perceived importance of the topic—affect responsibility attribution when agent-assisted decisions fail. To test this, we formulated two hypotheses and conducted a factorial experiment involving three potential responsibility targets: the user, the agent, and the agent's developer or service provider.

The findings partially supported **H1**. A significant interaction was observed between prior knowledge and responsibility target, consistent with the statistical results reported earlier. Simple main effects analysis showed that when participants lacked prior knowledge of the agent, they assigned significantly more responsibility to the user. In contrast, participants with prior knowledge tended to assign more responsibility to the agent itself. However, the effect of prior knowledge on developer responsibility did not reach conventional significance, and therefore H1 should be interpreted as being only partially supported. These results suggest that prior exposure to an agent's capabilities or personality can shift accountability away from the human user and toward the system.

The results also supported **H2**. The interaction between topic importance and responsibility target was significant. While no clear differences emerged under low- or medium-importance conditions, responsibility attribution shifted markedly under high-importance conditions. In particular, participants assigned the highest level of responsibility to the developer or service provider, followed by the agent, and the lowest to the user. This indicates that when the stakes are high, people are more likely to hold the system's creators accountable—potentially reflecting stronger expectations for reliability and oversight in serious contexts. Together, these results validate both hypotheses and highlight the nuanced ways in which people assign blame in human-agent interactions.

Beyond the empirical findings, this study contributes theoretically to our understanding of responsibility attribution in human-agent interaction. Prior work has extensively examined factors such as autonomy, transparency, and anthropomorphism, but has often treated responsibility as a static construct. Our results demonstrate that responsibility attribution is dynamic, shaped both by users' cognitive framing (e.g., prior knowledge of the agent) and by contextual framing (e.g., the perceived importance of the decision). This dual modulation provides empirical grounding for a dynamic model of blame attribution, in which user expectations and situational seriousness jointly determine how accountability is distributed. This insight is particularly relevant for system designers and policymakers, who must anticipate shifts in responsibility attribution depending on user familiarity with agents and the gravity of the situation at hand.

Beyond responsibility attribution, these findings also relate to broader discussions in explainable and human-in-the-loop AI. Recent work on AI validation frameworks⁵³, user-centered explanation design⁵⁴, and end-user explanation needs⁵⁵ highlights that users form expectations not only from system performance but also from how agents communicate their capabilities and limitations. These perspectives complement our results, suggesting that responsibility judgments may reflect how users question, interpret, and validate agent behavior. Integrating responsibility attribution with XAI-oriented design practices may therefore help create systems that better manage user expectations and reduce misaligned blame in real-world deployments.

These findings can also be situated within broader accountability theory. In many real-world deployments, humans often absorb disproportionate blame when automated systems fail—a phenomenon described by Elish⁵⁶ as the “moral crumple zone.” In contrast, our results show that providing prior knowledge of an agent can reverse this dynamic by shifting responsibility away from the human user and toward the system itself. This pattern also relates to debates on the “responsibility gap”⁴⁷, which highlight difficulties in assigning accountability when outcomes emerge from distributed human–AI interactions. Furthermore, the observed sensitivity to contextual seriousness aligns with discussions of “meaningful human control,” emphasizing that users and policymakers must anticipate when and how responsibility should be shared between humans and AI systems. Integrating these frameworks helps position our findings within the larger discourse on accountability in socio-technical systems.

Strengths and limitations

One strength of this study is its demonstration that providing prior knowledge of an agent can increase the extent to which both the agent and its developer or provider are held accountable. This effect may be a consequence of framing the agent positively—emphasizing its abilities and successful behavior—in order to encourage trust and adoption. In this experiment, participants in the prior knowledge condition viewed a self-introduction video (Video 1) in which the agent's capabilities were presented in a successful context, while the main task video (Video 2) involved a failure scenario. This design mirrors real-world contexts, where systems are often promoted on the basis of their benefits rather than their potential shortcomings.

However, the prior-knowledge manipulation may have influenced responsibility attribution not only through cognitive familiarity but also through increased anthropomorphism and perceived agency. Because the manipulation involved an agent's self-introduction video, participants may have construed the agent as more socially present or “mind-like,” which can systematically alter blame attribution^{10,12}. Thus, the observed knowledge effect may partially reflect a broader social framing process rather than purely informational understanding. Future work should therefore disentangle cognitive familiarity from anthropomorphic or social cues to clarify their respective contributions.

Also, the manipulation of topic importance also introduces a potential confound. In the present study, the low-, medium-, and high-importance conditions differed not only in perceived stakes but also in their narrative content (job notification, travel planning, and emergency evacuation). As a result, the scenarios may have varied in uncontrolled ways, including differences in emotional arousal, moral urgency, and perceived reversibility. Although the pretest confirmed that the intended importance levels were successfully differentiated, it did not establish equivalence across these additional affective or moral dimensions. Future work should therefore employ scenarios with constant content while manipulating consequences or framing alone, in order to isolate the pure effect of perceived importance.

At the same time, the results suggest that such promotional framing may have unintended consequences: it can lead users to hold agents and developers more responsible when failures occur. This has important implications for how AI systems are introduced and explained to users. With respect to topic importance, the study found that higher importance was associated with lower responsibility attributed to the user. This pattern suggests a psychological distancing effect: when the stakes are high, users may deflect blame and place greater responsibility on external agents or their creators, potentially as a means of reducing cognitive dissonance or perceived personal fault.

A key limitation of this study is that participants were not active users of the agent during the experiment. Rather, they observed a video of a user–agent interaction. This design choice was made to avoid imposing discomfort or disadvantage on participants, as the experiment involved assigning blame in a failure scenario. However, this limits ecological validity, as responsibility attribution may differ when individuals are directly

engaged in the interaction or decision-making process. Future research should therefore investigate responsibility attribution in more immersive or personally relevant contexts—for example, by asking participants to reflect on past interactions with AI, or by embedding agents into real-time decision-making tasks. Such approaches may offer deeper insights into how direct involvement shapes accountability, and how responsibility trade-offs emerge in applied domains such as healthcare, mobility, and education.

In addition, the sample characteristics introduce cultural considerations. All participants were Japanese adults recruited from Yahoo! Crowdsourcing, with a mean age of approximately 50. Cultural norms surrounding blame, responsibility, and trust in technology are known to vary cross-nationally, which may limit the generalizability of these findings. Furthermore, because participants evaluated responsibility as observers rather than as direct users, the present results reflect vicarious responsibility attributions rather than first-person judgments. Although this does not undermine the internal validity of the findings, it narrows the scope of their application. Future studies should examine whether similar patterns emerge in cross-cultural samples and in settings where participants make decisions themselves.

Conclusion

As humans and agents increasingly coexist in society, questions of responsibility attribution—particularly toward agents and their developers or providers—are critical to the acceptance and ethical deployment of agent-based systems. This study investigated how responsibility is assigned when agent-supported decision-making results in failure, using a three-factor mixed design that examined prior knowledge of the agent, topic importance, and the responsibility target. The results revealed two key findings: (1) a lack of prior knowledge of the agent led participants to assign greater responsibility to the user, and (2) as the importance of the topic increased, responsibility shifted more strongly toward the agent's developer or provider. These findings supported both of our initial hypotheses. Overall, the study provides empirical evidence that perceptions of accountability in human–agent interaction are shaped by contextual and cognitive factors. These insights offer valuable guidance for the design and communication of agent systems—particularly in understanding when and why users may shift blame toward agents or those who create them. As such, this work contributes to the development of more socially attuned and ethically aware agents.

Data availability

Data is provided within the manuscript or supplementary information files

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Author contributions

TT contributed to the conception, design, and execution of the study, performed the statistical analysis, and wrote the first draft and final version of the manuscript. SY served as the driving force behind the concept, organized the project, and provided guidance throughout the execution of the project. All authors contributed to manuscript revision and read and approved the submitted version.

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Declarations

Competing interests

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Additional information

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